



Short-period surface-wave phase velocities across the conterminous United States

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ABSTRACT

Surface-wave phase-velocity maps for the full footprint of the USArray Transportable Array (TA) across the conterminous United States are developed and tested. Three-component, long-period continuous seismograms recorded on more than 1800 seismometers, most of which were deployed for 18 months or longer, are processed using a noise cross-correlation technique to derive inter-station Love and Rayleigh dispersion curves at periods between 5 and 40 s. The phase-velocity measurements are quality controlled using an automated algorithm and then used in inversions for Love and Rayleigh phase-velocity models at discrete periods on a 0.25°-by-0.25° pixel grid. The robustness of the results is examined using comparisons of maps derived from subsets of the data. A winter–summer division of the cross-correlation data results in small model differences, indicating relatively minor sensitivity of the results to seasonal variations in the distribution of noise sources. Division of the dispersion data based on inter-station azimuth does not result in geographically coherent model differences, suggesting that azimuthal anisotropy at the regional scale is weak compared with variations in isotropic velocities and does not substantially influence the results for isotropic velocities. The phase-velocity maps and dispersion measurements are documented and made available as data products of the 10-year-long USArray TA deployment.

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1. Introduction

In early October, 2013, the final station of the USArray Transportable Array (TA) was installed in eastern Maine, to complete the TA footprint in the conterminous US. Approximately two years later, in late 2015, the stations of the easternmost and final swath of the initial TA deployment were removed. This completed the 10-year-long data-collection phase of the first TA experiment, with the second experiment now underway with station deployments across the state of Alaska.

The TA data set, covering the lower 48 states of the United States on a uniform grid of approximately ~70 km station separation and with nearly two-year-long station deployments, has proved remarkably rich, and has facilitated a wide range of seismological investigations of seismic sources, seismic wave propagation, Earth structure, and additional topics. Elucidating the structure and evolution of the North American continent was the main research focus put forward in the early USArray community-planning documents (Ekström et al., 1998; Levander et al., 1999; Meltzer et al., 1999) and this objective was reiterated and expanded on in the 2010 EarthScope Science Plan (Williams

et al., 2010). The number of articles exploiting the USArray TA data set to this end has grown rapidly during the progression of the array across the continent. A 2014 special issue of *Earth and Planetary Science Letters* (Long et al., 2014) provides a good sample of the research sparked by the USArray TA, and many of the studies reported there and other investigations are ongoing.

While a primary seismological goal of USArray is a fully 3-D description of the elastic and anelastic structure of the North American continent, it has been recognized since the early days of planning the project that intermediate research results in the form of well-documented measurements and less-complex models, such as 2-D phase-velocity maps and 1-D local velocity profiles, are of great value (e.g., IRIS, 2005). The importance of these ‘data products’ is that, while there may be disagreements between researchers about the best way to parameterize and derive a 3-D model of the Earth, agreement may be found around underlying observations and simpler parameterized models derived from well-documented observations. Agreement between data products provides a level of validation of data and methods that is useful for supporting the adoption of large and diverse observations and low-level models in the derivation of more complex 3-D models.

In this paper, I report on the completion of an investigation of short-period Love and Rayleigh wave propagation across the footprint of the TA, and present the corresponding data products,

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consisting of dispersion measurements and isotropic phase-velocity maps. The work presented here is an extension of research based on TA data recorded through 2012 described in an earlier study (Ekström, 2014). Preliminary data products were presented in the earlier paper. The analysis and data products presented here include three additional years of TA data, mainly from stations in the easternmost US as well as in southeastern Canada.

The technique used in this paper to derive surface-wave phase-velocity measurements is based on the cross-correlation of continuous seismic noise recorded on a pair of stations. The noise cross-correlation approach, pioneered in seismology by Aki (1957), has been implemented in a number of algorithms by different authors (e.g. Shapiro et al., 2005; Bensen et al., 2007), and has been applied successfully in dozens of papers to subsets of the TA data set (e.g. Lin et al., 2008). The algorithm used in this study has been described previously (Ekström et al., 2009; Ekström, 2014), and its relationship to other implementations is also discussed in these earlier papers.

The focus of the current paper is a new, complete set of Love and Rayleigh phase-velocity maps for the conterminous US derived using the approach described by Ekström (2014). For completeness, the key elements of the methods are summarized in Section 2 of the paper. Section 3 presents the data selection and the results of the inversions for isotropic phase-velocity maps. In Section 4, I present and discuss results from three experiments aimed at exploring the robustness of the phase-velocity maps, and especially the biases that may exist as the result of unaccounted-for complexity of the background noise field and of unmodeled azimuthal anisotropy of the Earth. Section 5 provides a description of the data products resulting from this research, and conclusions.

2. Summary of methods

The processing and analysis occurs in four steps. In the first step, continuous long-period seismograms from the USArray TA and additional stations are cross-correlated in 4-h-long time segments to form normalized cross-correlation spectra $\rho_{ijk}(\omega)$, where i and j are station indices and k is a time-segment index. The normalized cross-correlation spectra are summed to form stacked spectra,

$$\rho_{ij}^S(\omega) = \sum_{k=1}^{N_{ij}} \rho_{ijk}(\omega), \quad (1)$$

where N_{ij} is the total number of 4-h-long records available for a station pair (typically around 3000), and $\rho_{ij}^S(\omega)$ is the stack.

The second and third steps of the analysis build on the spectral approach of Aki (1957). In his formulation, the observed cross-correlation spectra are compared with the theoretical expression for the cross correlation of data from two stations in an isotropic surface-wave noise field,

$$\bar{\rho}(r, \omega_0) = J_0\left(\frac{\omega_0}{c(\omega_0)}r\right), \quad (2)$$

where $\bar{\rho}(r, \omega_0)$ is the cross correlation, r is the receiver separation, J_0 is the Bessel function of the first kind and zeroth order, and $c(\omega_0)$ is the phase velocity between the receivers at frequency ω_0 . In the second step, the zero crossings of the real part of the stacked cross-correlation spectra are identified using a simple algorithm. In the third step, the observed zero crossings are associated with specific zeros of the Bessel function, leading to corresponding phase-velocity estimates,

$$c(\omega_n^Z) = \frac{\omega_n^Z r}{Z_m}, \quad (3)$$

where z_m is the m th zero of the Bessel function, and ω_n^Z is the frequency of the n th observed zero of the cross-correlation function. Most of the complexity of the algorithm is related to making correct associations of observed zeros with zeros of the Bessel function and eliminating unreliable observations. Once a sequence of zeros has been associated, the discrete $c(\omega_n^Z)$ are joined to define a dispersion curve.

In a small modification to the approach of Ekström (2014), in which Eq. (2) was used for determining phase velocities associated with zero crossings not only for the vertical component but also for the transverse component, I here instead use the corresponding expression derived for Love and Rayleigh wave motion recorded on horizontal components (Aki, 1957; Haney et al., 2012),

$$\bar{\rho}_H(r, \omega_0) = \frac{1}{2}J_0\left(\frac{\omega_0}{c(\omega_0)}r\right) - \frac{1}{2}J_2\left(\frac{\omega_0}{c(\omega_0)}r\right), \quad (4)$$

where $\bar{\rho}_H$ denotes a horizontal-component cross-correlation function, and J_2 is the Bessel function of the first kind and second order. As $-J_2$ asymptotically approaches J_0 for large values of the argument, this modification is not expected to have a large effect on the results. As discussed below, maps obtained using Eqs. (2) and (4) are, indeed, nearly indistinguishable.

In the fourth step of the analysis, the dispersion curve is interpolated at discrete periods and the phase velocity is rewritten as an observed travel time, $\tau_{ij} = \frac{X_{ij}}{c_{ij}}$, where X_{ij} is the distance between stations i and j , c_{ij} is the measured phase velocity, and τ_{ij} is the corresponding travel time. The travel times are then used to map the geographic variations of isotropic phase slowness across the footprint of USArray using a parameterization in terms of N latitude-longitude pixels. The model travel times τ_{ij}^p are constructed as

$$\tau_{ij}^p = \sum_n X_{ij}^n p_n, \quad (5)$$

where the slowness in each pixel n is p_n , and the fractional path length in each pixel is X_{ij}^n . The data fit is calculated as

$$\chi^2 = \sum_{ij}^K \left(\frac{\tau_{ij} - \tau_{ij}^p}{\sigma_{ij}} \right)^2, \quad (6)$$

where the summation is over the K station pairs ij for which an observation is included at a particular period, and σ_{ij} is an estimate of the observational uncertainty. The best-fitting slowness model is determined by solving the damped least-squares problem

$$\min(\chi^2 + \nu R^2), \quad (7)$$

where R^2 is the roughness of the slowness variations and ν is a smoothing coefficient.

3. Data and results

Data recorded on the TA stations for the period January 2006 through December 2015 were collected and processed. The set of TA stations was augmented by stations of the US National Seismic Network (US), the Caltech Regional Seismic Network (CI), the Berkeley Digital Seismograph Network (BK), the ANZA Regional Network (AZ), the Leo Brady Network (LB), the Western Great Basin/Eastern Sierra Nevada Network (NN), the Lamont-Doherty Cooperative Seismographic Network (LD), and the Global Seismographic Network (IU and II). Data from 33 stations of the Canadian Polaris Network (PO) operating in southeastern Canada were also included. Cross-correlation stacks were calculated for vertical, transverse, and radial components of all station pairs separated by less than 600 km. Dispersion curves were determined and sampled for inter-station travel times at 5, 6, 8, 10, 12, 15, 20, 25, 30,

Table 1
Data set and inversion results. Period (T) and the distance range included ($X_{\min}$ and $X_{\max}$), the number of observations (N) used in the inversion, the average phase velocity (c) and the variance reduction (ζ) at each period and for each wave type. The variance reduction is calculated with respect to the best-fitting uniform-velocity model at each period, and includes outliers: $\zeta = \sum (\Delta\tau - \Delta\tau^p)^2 / \sum (\Delta\tau)^2$, where $\Delta\tau$ is the observed residual, and $\Delta\tau^p$ is the predicted residual.

T s	$X_{\min}$ km	$X_{\max}$ km	Love (transverse)			Rayleigh (vertical)			Rayleigh (radial)		
			N	c km s ⁻¹	ζ	N	c km s ⁻¹	ζ	N	c km s ⁻¹	ζ
5	40	200	28,577	3.11	0.95	28,260	2.98	0.93	25,285	2.97	0.90
6	48	200	31,082	3.25	0.95	31,618	3.05	0.93	27,973	3.04	0.89
8	64	200	31,788	3.43	0.95	32,774	3.15	0.94	28,193	3.14	0.88
10	80	200	29,123	3.54	0.94	29,636	3.21	0.93	22,947	3.21	0.88
12	96	240	44,117	3.62	0.92	44,416	3.28	0.90	28,376	3.27	0.88
15	120	300	65,936	3.72	0.90	70,158	3.37	0.88	42,836	3.37	0.81
20	160	400	64,583	3.86	0.85	92,772	3.53	0.91	27,817	3.52	0.66
25	200	500	84,710	3.98	0.80	139,955	3.67	0.88	36,527	3.66	0.64
30	240	600	100,564	4.08	0.77	206,285	3.77	0.87	39,100	3.77	0.69
35	280	600	84,722	4.17	0.72	188,453	3.85	0.84	27,617	3.85	0.69
40	320	600	65,862	4.25	0.66	161,253	3.90	0.81	17,061	3.91	0.66

35, and 40 s, leading to approximately 3.5 million measurements on each component.

The data set was trimmed before inversion to include, at each period, measurements corresponding to path lengths of a few (~ 2 – 5) wavelengths. The specific minimum and maximum distances ($X_{\min}$ and $X_{\max}$) used for the selection at each period are given in Table 1. Data from 1847 stations were used in the determination of the final models. A two-step inversion, with an initial inversion parameterized on a coarser 0.5° -by- 0.5° pixel grid and a

second inversion on the final 0.25° -by- 0.25° pixel grid, was used to identify outlying observations and reduce their influence on the final models. Details of the inversion can be found in Ekström (2014).

The Love and Rayleigh slowness variations that are imaged by solving the minimization problem (Eq. (7)) are the main results and data products of this study. Love wave maps were obtained from the transverse-component dispersion data. Rayleigh wave maps were obtained separately from the vertical- and radial-

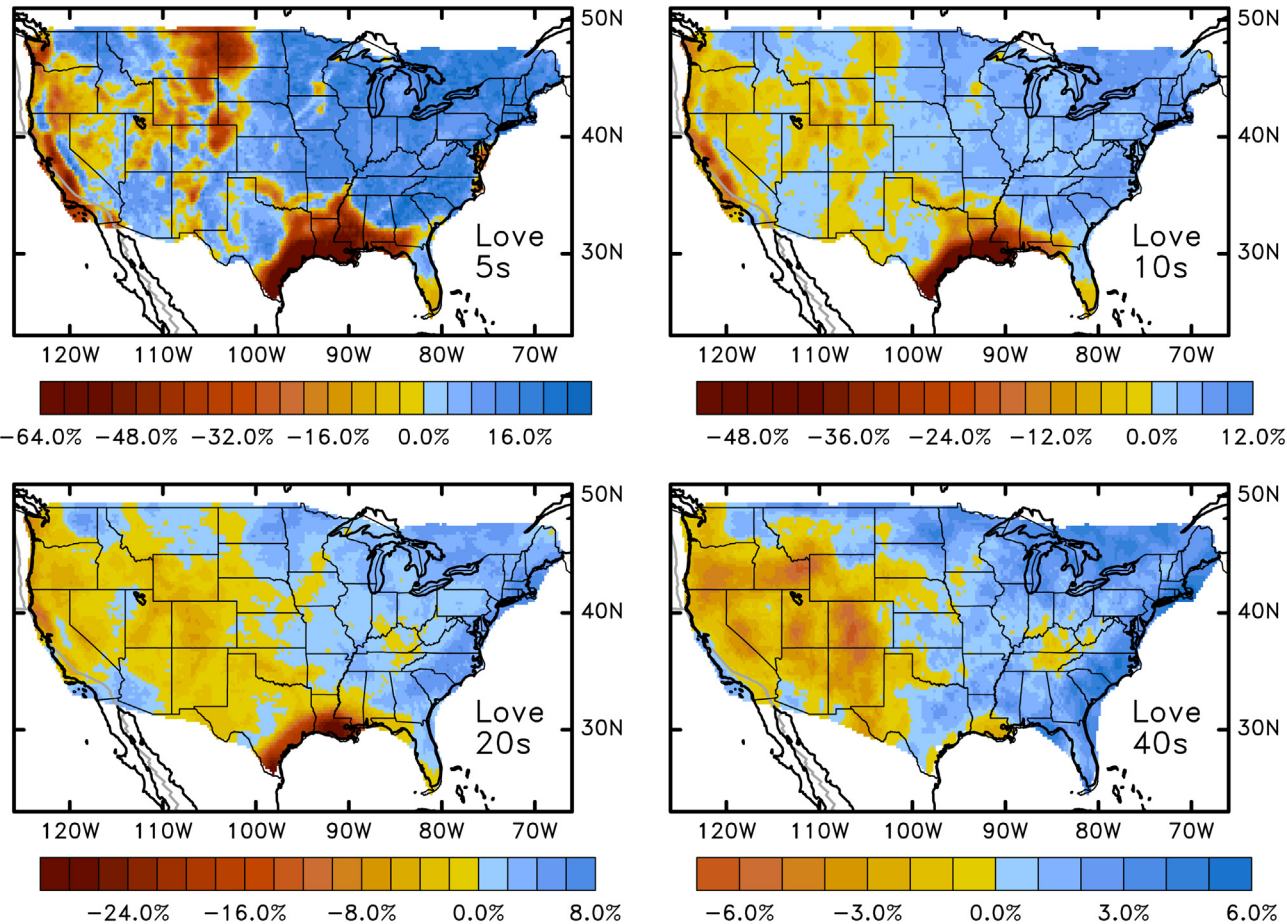


Fig. 1. Maps of Love wave phase velocity at different periods: top left, 5 s; top right, 10 s; bottom left, 20 s; and bottom right, 40 s. The maps are parameterized using 0.25° -by- 0.25° pixels, and deviations are shown with respect to the mean (Table 1). Note the different scales at different periods, and the skewed distribution of deviations at short periods.

component dispersion data, allowing for a comparison of the two sets of results. In the presentation and discussion of the maps below, the phase slowness was converted to phase velocity. Table 1 summarizes the number of data used, average velocities, and reduction in data variance provided by the maps. Figs. 1 and 2 show the preferred results for Love and Rayleigh wave phase-velocity variations at 5 s, 10 s, 20 s and 40 s.

4. Robustness of the phase-velocity models

The variance reduction achieved is very good for all periods analyzed, for both Rayleigh and Love waves (Table 1). However, the derived phase-velocity maps are not perfect representations of true structure. The data misfit reflects errors and inconsistencies in the observations, some of which are mapped into the models. Uncertainties related to assumptions inherent in the noise cross-correlation technique and to the ray-theoretical behavior assumed in the modeling also influence the final maps, as do the choices of model parameterization and regularization. To explore the robustness of and uncertainties in the preferred maps, I compare results obtained from analyses of subsets of the observations.

4.1. Horizontal-component phase-velocity maps

Studies of surface-wave dispersion based on the cross-correlation of noise commonly focus on signals recorded on the vertical component, from which Rayleigh wave dispersion is

extracted by application of Eq. (2), or using other methods. Rayleigh wave dispersion measurements can also be obtained from the cross-correlation of radially polarized horizontal components, and Love wave dispersion is obtained from the transversely polarized horizontal components. Aki (1957) described how the interference of horizontal wave motion differs from that of vertical wave motion, and how these differences lead to the relationship described in Eq. (4). The zero crossings of the difference $J_0 - J_2$ are shifted to smaller arguments with respect to those of J_0 , which results in faster phase velocities than would be obtained using Eq. (2). However, the effect is small except for very small station separations, since zero crossings of J_2 quickly approach those of J_0 for arguments greater than $4\pi(2\lambda)$.

In practice, for the approach used here, Rayleigh and Love wave maps determined from measurements derived using the two equations are nearly indistinguishable, with correlation coefficients between maps at all periods greater than 0.998. An anticipated, but small, effect is observed on the mean phase speeds, with the average phase velocities being 0.03–0.3% faster for models determined from horizontal data using Eq. (4) as compared with Eq. (2).

A comparison of the results for Rayleigh wave dispersion measured on the vertical and radial components is shown in Fig. 3. The top panels show the Rayleigh wave phase-velocity maps determined at 5 s and 25 s from the radial-component dispersion measurements, and the bottom panels show the difference maps with respect to the corresponding models determined from the vertical-component measurements. At all periods, the maps derived from

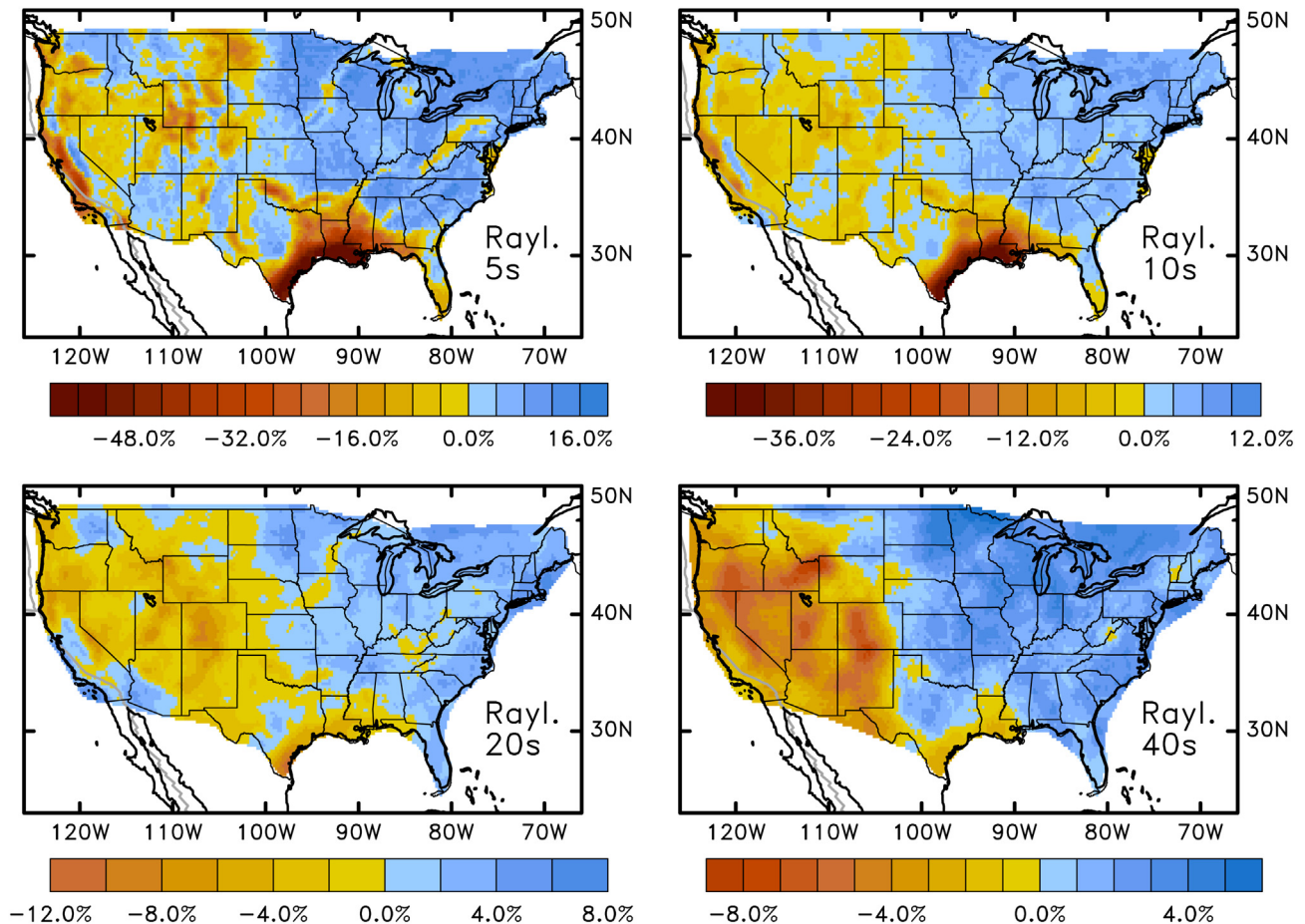


Fig. 2. Maps of Rayleigh wave phase velocity determined from the vertical component at different periods: top left, 5 s; top right, 10 s; bottom left, 20 s; and bottom right, 40 s. The maps are parameterized using 0.25°-by-0.25° pixels, and deviations are shown with respect to the mean (Table 1). Note the different scales at different periods, and the weaker expression of some short-period anomalies as compared with the Love wave maps (Fig. 1).

the vertical- and radial-component data sets are very similar, with correlation coefficients greater than 0.94 for all periods, and greater than 0.98 for periods shorter than 15 s. Average phase velocities differ by 0.5% for the shortest-period maps (5 and 6 s), and by less than 0.2% for all other periods. The high level of similarity between results derived from different data sets gives an indication of the robustness of the results. Since the vertical-component dispersion data set is larger than the radial-component data set, and is better fit, we prefer the vertical-component results.

The similarity of Rayleigh wave phase-velocity maps derived from vertical- and horizontal-component data also has implications for the Love wave results. As the numbers in Table 1 show, the Love wave data are more numerous than the radial-component Rayleigh wave data at all periods, and are also better fit. This suggests that the Love wave results are of comparable quality to the Rayleigh wave results.

4.2. The effect of seasonal variations of noise

One of the main assumptions in the derivation of Eqs. (2) and (4) is that the azimuthal distribution of noise sources is isotropic. Since it is generally accepted that most of the ‘noise’ used in noise cross-correlation studies such as this one has its source in the ocean, there is no reason to believe that the isotropic condition is met (e.g. Yao and van der Hilst, 2009; Tsai, 2009; Harmon et al., 2010). Instead, what is often asserted is that the distribution of noise sources is isotropic enough for the method to work.

To investigate the influence of the non-isotropic distribution of noise sources, I compare results obtained from observations during the winter months (November–April) with those obtained from observations during the summer months (May–October). Since the strength of wave action in the ocean is highly correlated with hemispheric winter, the distribution of noise sources during winter and summer is expected to be different, which could potentially lead to bias in the phase-velocity maps.

Analyses were performed on stacked seasonal subsets of cross-correlations, using the same selection criteria and inversion parameters as in the analysis of the full data set. Remarkably, there are hardly any differences between the maps derived from summer data and from winter data. For Rayleigh waves, the correlations between summer and winter models are all greater than 0.98. For Love waves, the correlations are all greater than 0.93 and are greater than 0.98 for periods shorter than 25 s. Differences in the average phase velocity are 0.1%, or smaller. Fig. 4 shows examples for Rayleigh waves at 10-s period and Love waves at 20-s period. Model differences apparent along the Gulf Coast for the longer-period Love wave maps are mainly differences in anomaly intensity, with the very low velocities being imaged as even slower using the winter cross-correlation data.

4.3. The effect of azimuthal selection

In this study, I have chosen to parameterize the phase-velocity variations across the conterminous US solely in terms of isotropic velocities. By not including azimuthal velocity variation in the

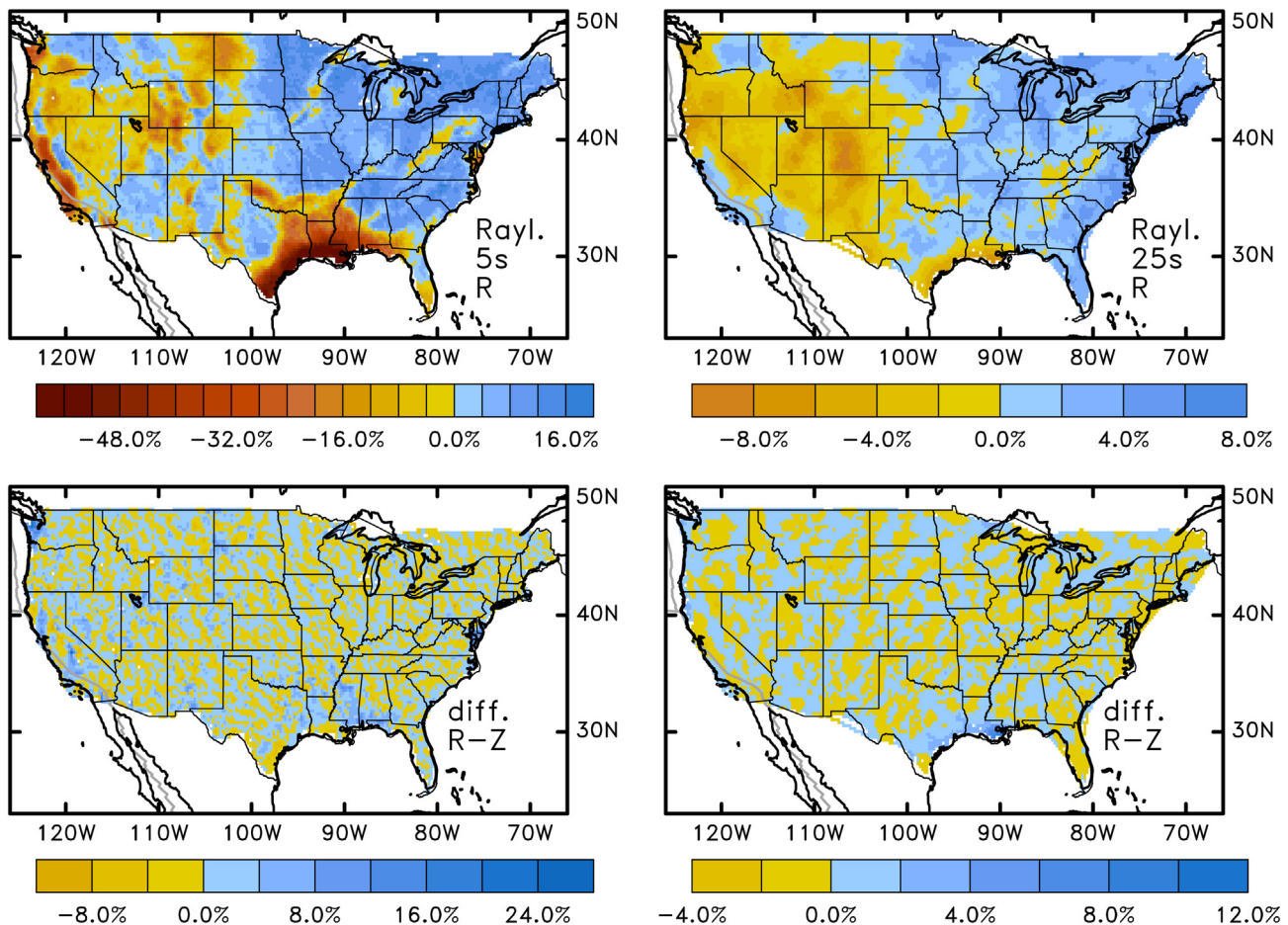


Fig. 3. (top panels) Maps of Rayleigh wave phase velocity derived from radial-component (R) measurements at different periods: top left, 5 s; top right, 25 s; (bottom panels) Maps showing the difference between models derived from vertical-component (Z) data and the models shown in the top panels: bottom left, 5 s; bottom right, 25 s. The maps are parameterized using 0.25°-by-0.25° pixels.

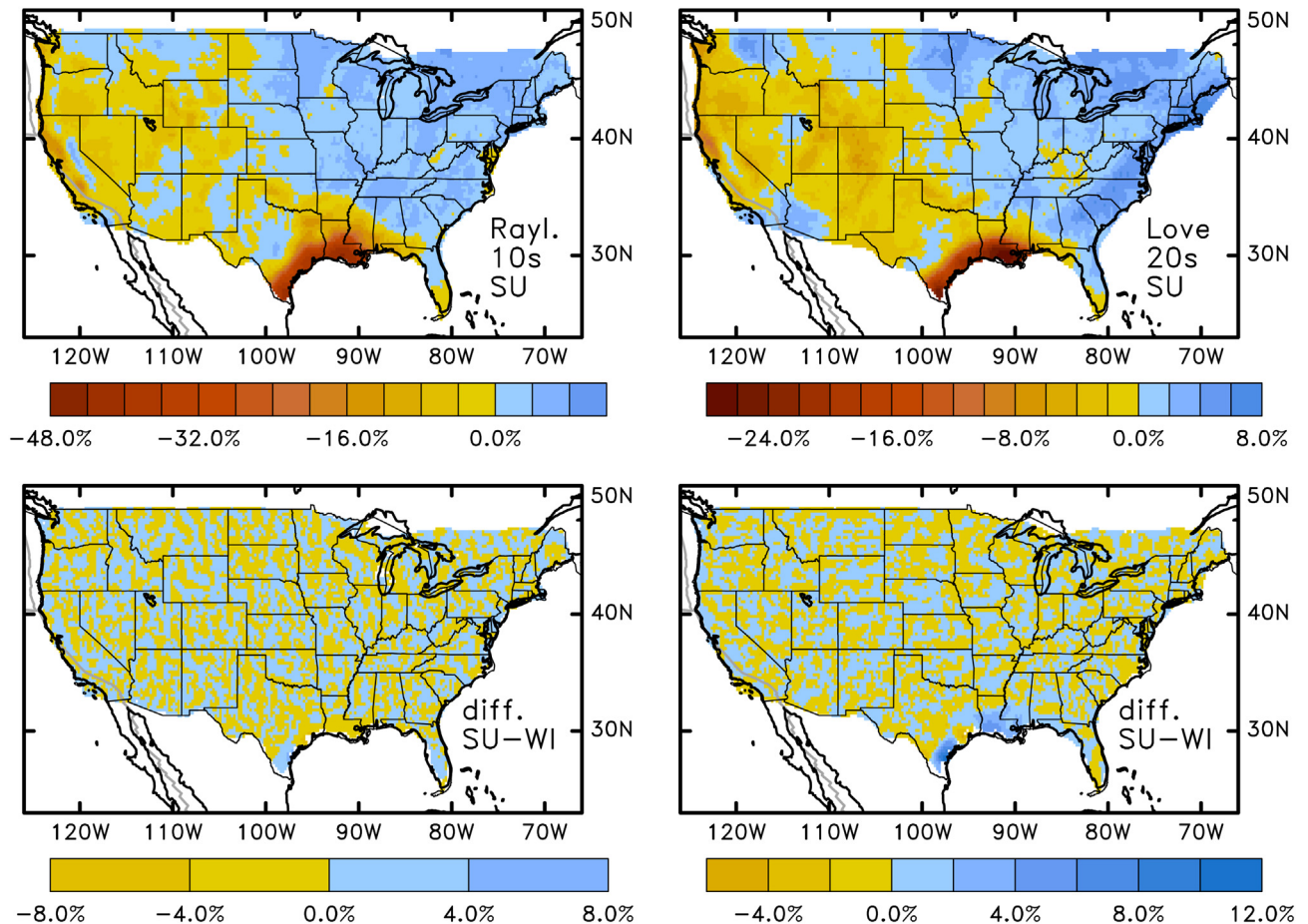


Fig. 4. (top panels) Maps of phase velocity derived from data recorded during summer (SU) months: left, 10-s Rayleigh waves; right 20-s Love waves. (bottom panels) Maps showing the difference between models derived from summer and winter (WI) data (summer map minus winter map).

parameterization, the inversion is simplified, and consideration of potentially complicated trade-offs between isotropic and anisotropic variations is avoided. However, elastic anisotropy is known to be prevalent in crustal and shallow-mantle rocks. Here, I investigate the extent to which existing seismic anisotropy may lead to bias in the isotropic phase-velocity maps derived in this study.

Intrinsic anisotropy is reflected in surface-wave phase velocities as $\cos 2(\varphi - \alpha)$ and $\cos 4(\varphi - \beta)$ directional variations, where φ is the azimuth of the path of propagation, and α and β are fast directions. In general terms, the nearly uniform station distribution of the TA is favorable for achieving the uniform azimuthal coverage needed to average out azimuth-dependent surface-wave phase-velocity variations in the inversion. However, a path selection with a directional bias could bias the isotropic velocities retrieved in the final models. Such a path selection might arise from the combined effect of an uneven distribution of noise sources and preferential selection of inter-station paths with clear cross-correlation signals.

To investigate the potential effects on the isotropic models of azimuthal anisotropy and path distribution, I divide the inter-station dispersion data into subsets based on azimuth, each containing approximately half of the observations of the full data set. The four subsets contain paths whose azimuths φ or $\varphi - 180^\circ$ fall in different azimuthal quadrants: 0° – 90° , 45° – 135° , 90° – 180° , and 135° – 225° . The first and third and the second and fourth subsets have no overlapping observations. Other combinations of subsets will have half of their observations in common.

Fig. 5 shows two example maps of Love and Rayleigh waves at 6-s period derived from the N-S data set (the fourth data subset), and corresponding difference maps with respect to the E-W data set (the second subset). For both Love and Rayleigh waves at this period, the correlations between the maps derived from N-S data and E-W data are 0.97, and the average velocities are different by less than 0.1%. The difference maps in Fig. 5 show that most of the differences between the models appear geographically random, with a few exceptions. In particular, the N-S trending linear anomalies associated with coastal mountain ranges in California, Oregon and Washington are imaged slightly differently by the two data sets. I attribute most of these localized differences to the limited resolution afforded by the E-W data set for imaging narrow N-S structures.

Pairwise correlations of maps derived from the four subsets of data at all periods (5–40 s) are generally high, and no significant variation is observed in the mapped mean velocities. Specifically, for Rayleigh waves at all periods, the correlation of any two models is 0.96 or greater, and no two maps differ in their average velocity by more than 0.2%. For Love waves with periods shorter than 25 s, equally high correlation values and similarities in the mean velocities are found. For a few of the longer-period Love wave maps, correlations dip to 0.90 and differences in the means reach 0.4%. Visual inspection of these less-well correlated maps suggests that the variations are likely to be the result of insufficient regularization of the inversions given the smaller size of the subset of the data.

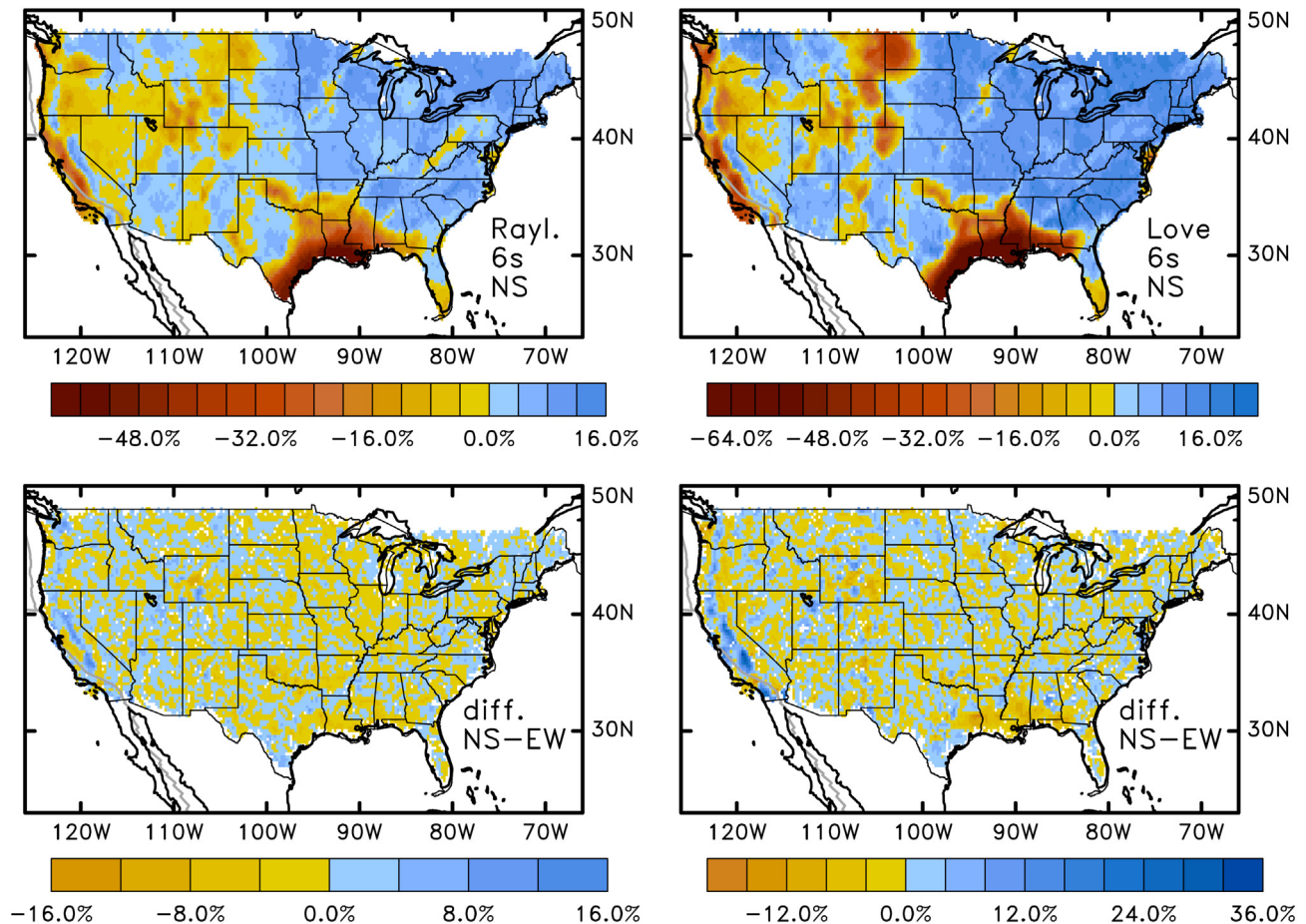


Fig. 5. (top panels) Maps of 6-s phase velocity derived from measurements of North–South (NS) inter-station paths: top left, Rayleigh waves; top right, Love waves. (bottom panels) Maps showing the difference between the models in the top panels and models derived from measurements of East–West (EW) inter-station paths. The maps are parameterized using 0.25°-by-0.25° pixels.

5. Conclusions

In this study, I have used continuous data from the 10-year deployment of the USArray TA across the conterminous US to measure short-period inter-station Love and Rayleigh phase velocities, and to map Love and Rayleigh surface-wave velocities across the TA footprint. The main contributions here are the systematic application of the automated measurement techniques developed in Ekström (2014) to the very large data set collected by the TA, and the derivation and testing of the corresponding phase-velocity maps.

Three experiments were performed to examine the robustness of the phase-velocity maps. In the first experiment, a comparison of Rayleigh wave maps determined independently from vertical- and horizontal-component dispersion data shows very small differences. The similarity of the maps illustrates the internal consistency of the vertical- and horizontal-component dispersion data. Considering the much smaller number of horizontal-component than vertical-component observations, presumably reflecting higher levels of local horizontal noise, the similarity also illustrates the overdetermined character of the inverse problem and the robustness of the isotropic phase-velocity maps.

The second experiment examined the model differences that result from dividing the cross-correlation data sets into two independent subsets based on time of year. Possible outcomes of this experiment were apparent model differences related to variations in the directionality of the Love and Rayleigh noise fields, or a bias

in the average velocities. Neither was observed. While it is likely that, in detail, some of the model differences can be traced to the distribution of noise sources, the dominant inference from this experiment is that the phase-velocity maps are robust with respect to real variations in the distribution of noise sources.

In the third experiment, I divided the dispersion measurements into subsets based on the direction in which they sample structure, determined phase-velocity maps from each subset, and compared the resulting maps. A potential outcome of this experiment was that imaging using half of the available azimuths would lead to unstable model results, or to model characteristics frequently attributed to uneven path distributions, such as the streaking of anomalies. A second potential outcome was noticeable effects of anisotropic structure on the imaged isotropic velocities due to preferential sampling in fast or slow directions. The latter would manifest itself as regionally coherent fast or slow areas in maps constructed from perpendicular subsets of paths, and potentially in different average model velocities, in the case of a dominant azimuthal fabric across the continent. Neither outcome is evident in the map differences or in the model averages, again illustrating the robustness of the results.

The implication of the third experiment is that the true level of underlying azimuthal anisotropy, combined with the level of preferential sampling in this study, is unlikely to affect the mapping of isotropic velocities significantly. In practical terms, it seems unlikely that azimuthal-anisotropy effects would be larger than the differences obtained between models derived from perpendicular

data sets. Here, these differences were shown to be small, in particular for Rayleigh waves, which typically are found to exhibit stronger azimuthal-anisotropy variations than Love waves. The implication of the experiment results is not that azimuthal anisotropy is insignificant or that it is not present in the dispersion data, but that azimuthal anisotropy is unlikely to bias significantly the isotropic maps derived here.

The phase-velocity models presented here have a direct application in predicting surface-wave propagation across the US. By combining the velocities at different periods, point-to-point dispersion curves can be constructed and used in the synthesis of short-period surface-wave seismograms. The main application for these results intended by the author is, however, as robust and geographically uniform constraints on the elastic structure of the crust and uppermost mantle of the conterminous US. This objective will be pursued in a future publication.

5.1. Models and data products

The preferred phase-velocity maps determined here are referred to collectively as model USANT15. The data sets and models made available as data products as part of this study include: (1) phase-velocity measurements made on vertical, radial, and transverse components for the full data set, and on the vertical and transverse component for the winter and summer seasonal data sets; (2) preferred phase-velocity maps of Love and Rayleigh waves; and (3) phase-velocity maps resulting from several of the experiments discussed in the text. The web address for the data products is <http://www.ideo.columbia.edu/~ekstrom/Projects/ANT/USANT15/>.

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